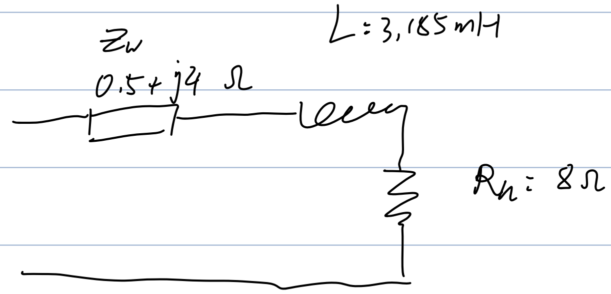
2024

Q1

(a)



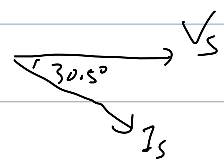
(i) $\omega = 2\pi f$:

$Z_n = j\omega L + R_n = 8 + 1j\ \Omega = 8.06\angle 7.1^\circ\ \Omega$

$Z_T = Z_n + Z_w = 8.5 + 5j\ \Omega = 9.86\angle 30.5^\circ\ \Omega$

(ii). $I_s = \frac{V_s}{Z_T} = 12.17\angle -30.5^\circ\ \Omega$

(iii).



(iv). $S = |V_s|\cdot|I_s| = 1460.4\ W$

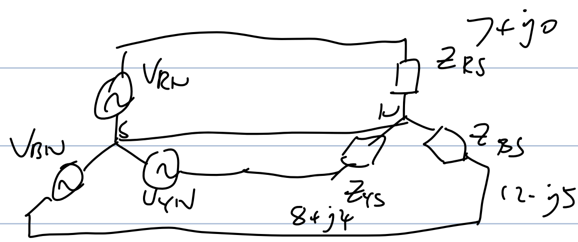
$Q = S\cdot\sin\theta = 741.2\ W$

$P = S\cdot\cos\theta = 1258.3\ W$

(v) $P_{out} = |I_s|^2\cdot R_n = 1184.9\ W$

$\text{efficiency} = \frac{P_{out}}{P} = 94.17\%$

(b),



$Z_{RS} = 7 + j0 = 7\angle 0^\circ\ \Omega$

$Z_{YS} = 8 + j4 = 8.94\angle 26.6^\circ\ \Omega$

$Z_{BS} = 12 - j5 = 13\angle -22.6^\circ\ \Omega$

Therefore:

$I_R = \dfrac{V_{RN}}{Z_{RS}} = \dfrac{6.6k/\sqrt{3}}{7\angle 0^\circ} = 544.36\angle 0^\circ\ A$

$I_Y = \dfrac{V_{YN}}{Z_{YS}} = \dfrac{6.6k\angle -120^\circ/\sqrt{3}}{8.94\angle 26.6^\circ} = 426.23\angle -146.6^\circ\ A$

$I_B = \dfrac{V_{BN}}{Z_{BS}} = \dfrac{6.6k\angle 120^\circ/\sqrt{3}}{13\angle -22.6^\circ} = 293.12\angle 142.6^\circ\ A$

(ii). $I_{SN} = I_R + I_Y + I_B = 71.90\angle -128.1^\circ\ A$

$V_{ph} = 6.6/\sqrt{3}\ kV = 3.81\ kV$

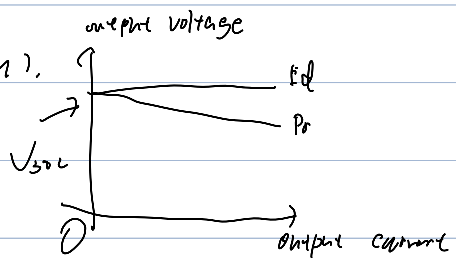
(iii) $\pi(\frac{d}{2})^2 = \dfrac{I_{SN}}{J} \Rightarrow d = 2\sqrt{\dfrac{I_{SN}}{\pi J}} = 55.2\ mm$

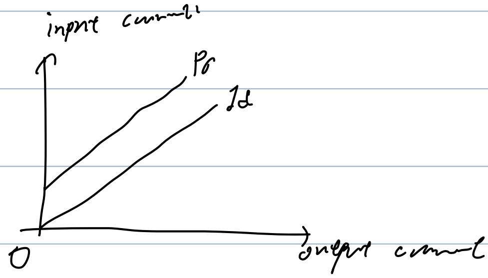
(iv) $P_{total} = |V_{ph}|\cdot|I_R|\cdot\cos\phi_1 + |V_{ph}|\cdot|I_Y|\cdot\cos\phi_2 + |V_{ph}|\cdot|I_B|\cdot\cos\phi_3$

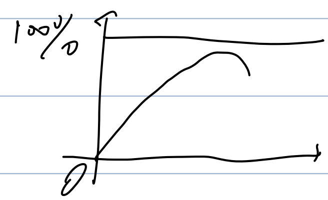
$= \dfrac{6.6k}{\sqrt{3}}\cdot\left(544.36\times\cos(0^\circ) + 426.23\times\cos(26.6^\circ) + 293.12\times\cos 22.6^\circ\right)$

$= 4.56\ MW$

Q2. (a).







(b). $V_{max} = \frac{2\pi}{\sqrt{2}} N \omega B_{max} A_c \approx 4.44 N \omega B_{max} A_c$

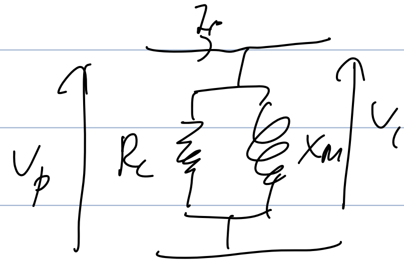
If overvoltage,

→ magnetic saturation : if > B_{max} → distorted current waveform → increase losses

→ maximum rms current

15:40 :

(c) In the open circuit :



$TR = \frac{V_p}{V_s} = \frac{240}{120} = 2$

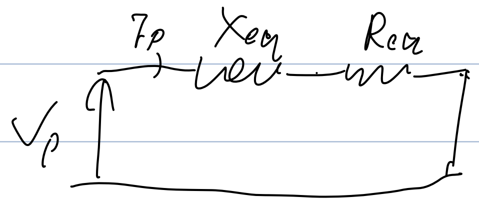
From $P_p = \frac{V_p^2}{R_c} \Rightarrow R_c = \frac{V_p^2}{P_p} = 720\ \Omega$

$S_p = I_p V_p = 144$ VA

$Q_p = \sqrt{S_p^2 - P_p^2} = 119.73$ VAr

From $Q_p = \frac{V_p^2}{X_m} \Rightarrow X_m = \frac{V_p^2}{Q_p} = 481.08\ \Omega$

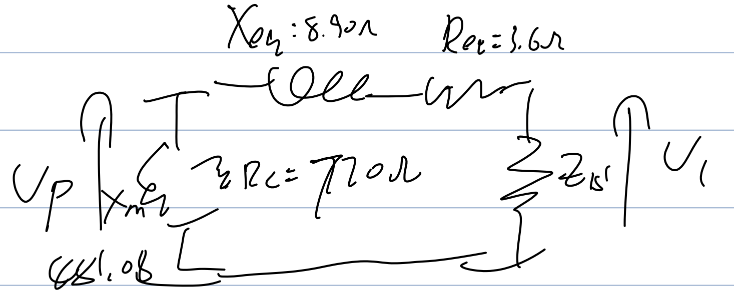
In the short circuit :



From $P_p = I_p^2 \cdot R_{eq} \Rightarrow R_{eq} = \frac{P_p}{I_p^2} = 3.6\ \Omega$.

$Q_p = \sqrt{(V_p I_p)^2 - P_p^2} = 222.49$ VAr

From $Q_p = I_p^2 X_{eq} \Rightarrow X_{eq} = \frac{Q_p}{I_p^2} = 8.90\ \Omega$



(d) (i). $Z_{LS}' = (TR)^2 \cdot Z_S = 12\ \Omega$

$Z_T = R_{eq} + Z_{LS}' + jX_{eq} = 15.6 + 8.9j\ \Omega = 17.96\angle 29.7^\circ\ \Omega$

$\Rightarrow I_p = \frac{V_p}{Z_T} = 13.36\angle -29.7^\circ\ A$

$V_1 = I_p \cdot Z_{LS}' = 160.32\ V$

$\Rightarrow V_2 = \frac{V_p}{TR} = \underline{80.16\ V}$

$\text{Regulation} = \frac{120 - 80.16}{120} \times 100\% = 33.2\%$

(ii) $P_{in} = |V_p| \cdot |I_p| \cdot \cos\theta = 2.785\ kW$

$P_{out} = \frac{V_2^2}{R_L} = 2.141\ kW$

$\text{efficiency} = \frac{P_{out}}{P_{in}} = 76.9\%$

(e). Let copper loss = iron loss, i.e.

$I_p^2 \cdot R_{eq} = \frac{V_p^2}{R_C} \Rightarrow I_p = \frac{V_p}{\sqrt{R_C R_{eq}}} = 4.71\ A$

$Z_T = \frac{V_p}{I_p} = 50.96\ \Omega$

$Z_T \approx Z_{eq} + Z_{LS}' \Rightarrow Z_{LS}' = 47.36\ \Omega$

$\Rightarrow Z_S = \frac{250.4}{TR^2} = 11.84\ \Omega$

$P_{out} = I_p^2 \cdot Z_{LS}' = 1.05\ kW$

$\text{loss} = I_p^2 \cdot R_{eq} + \frac{V_p^2}{R_C} = 159.86\ W$

$\Rightarrow \text{efficiency} = \frac{P_{out}}{P_{out} + \text{loss}} = 86.79\%$.

(a) (i). 3 single phase +

2 3phase +

Pro if one broken, others could still work

cons big. exp

Pros compact. effi

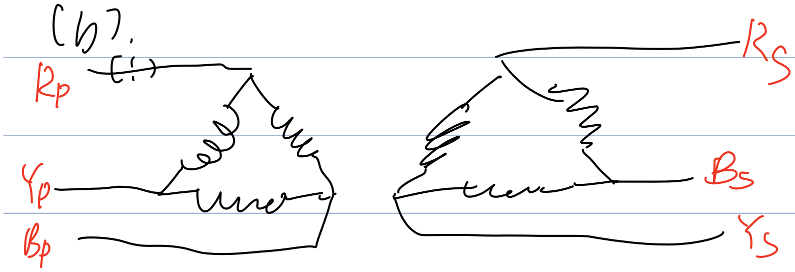
cons

only need to replace one transformer when faulty

smaller and cheaper

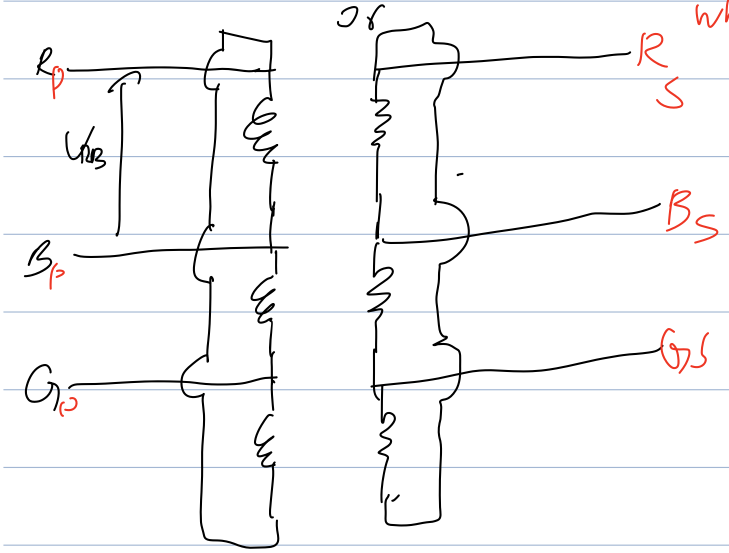
(b).

(i)



or

which one?



(ii). bank ratio $= V_{PL} / V_{SL} = 26.51$

phase ration $= V_{pph} / V_{sph} = 26.51$

(iii). From $S = 3 |V_{Lp}| \cdot |I_{pph}|$ Keep focus!

$\Rightarrow I_{pph} = \frac{S}{V_L} = 27.27 A / 3 = 9.09 A$

$\Rightarrow I_{pL} = \sqrt{3} I_{pph} = 47.24 A / 3 = 15.75 A$ I_{Lp}

The current in the primary coil is $I_{ph} = 27.77A / 3 = 9.29A$

$I_{SL} = I_{pL} \times \frac{11000}{415} = 1252.14\,A / 3 = 417.38\,A$

I_{phS}

$I_{sph} = \frac{I_{SL}}{\sqrt{3}} = 722.93\,A / 3 = 240.98\,A.$

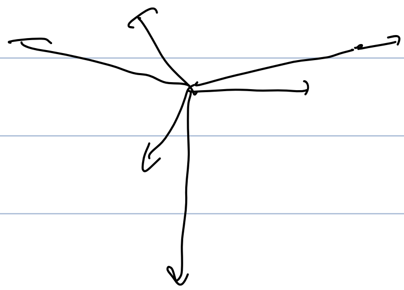
I_{Ls} I_{phS}

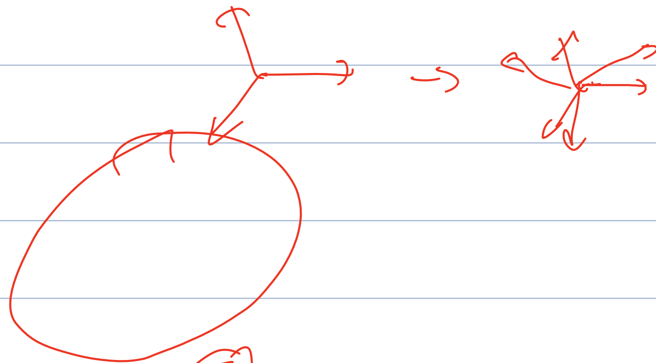
(iv) $I_{SL} = I_{sph} = 722.93A / 3 = 240.98A.$

$S = \sqrt{3}\,|V_{SL}| \cdot |I_{SL}| = 173.2\ kVA.$

Line or phase ?

(c).





$\sqrt{3}$?

bank ratio = 26.51

phase ratio = 45.91

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